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May 15, 1967
SUPERSEDING
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October 8, 1963

FEDERAL SPECIFICATION

DRILLS, ELECTRIC, PORTABLE (BATTERY-POWERED) AND ACCESSORIES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers battery powered portable electric drills, most generally used by the Federal Government, for drilling holes in wood, metal, and various other materials, in areas where electrical utility service is not available.

1.2 Classification.

1.2.1 Types and sizes. The portable electric drills shall be of the following types and sizes, as specified (see 6.1):

Type I—With cable assembly (battery adjacent to drill).

Type II—Without cable assembly (battery enclosed within drill).

Sizes (nominal chuck size)
¼ inch
⅜ inch
½ inch

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

W-C-265—Charger, Battery, Metallic Rectifier—Type (for Charging Nickel/Silver-Cadmium Power Packs).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-601—Boxes, Wood, Cleated—Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

MIL-I-43121—Interference Reduction for Electric Hand Tools.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

National Bureau of Standards (NBS) Handbook:

H28—Screw-Thread Standards for Federal Services.

(Copies of the National Bureau of Standards Handbook may be obtained upon application, accompanied by check, money order, coupon, or cash, to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C., 20402. Price may be obtained from the Superintendent of Documents).

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of drills which are otherwise in accordance with the requirements of this specification.

3.2 Material. The material used in the construction of the drills shall be new and shall be of good commercial quality which is suitable for the purpose intended. All materials used shall be free from defects and imperfections that may adversely affect the serviceability of the finished product. No material shall be used which constitutes a fire hazard or which requires special fire fighting techniques to extinguish.

3.3 Design. The drill shall have a low voltage (less than 30 volts) electric motor mounted in a suitable housing. The motor, powered by a battery source, shall transmit rotary motion to the drill chuck by means of a reduction gear mechanism.

3.4 Construction. Drills shall be simple in construction and reliable in operation. The parts shall be readily accessible for inspection, adjustment, or replacement. All parts of the drills shall be rugged in construction and supported, mounted and secured to withstand the abuses to which they may be subjected during normal usage. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable without requiring modification. All drills shall conform to the requirements of this specification and shall be capable of withstanding the tests specified in 4.5 and 4.6.

3.4.1 Fastening devices. All fastening devices, such as screws and pins, shall be installed with means for preventing the loss of tightness and, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.2 Threads. Threads shall conform to the applicable requirements of the National Bureau of Standards Handbook H28.

3.4.3 Finish. All drills shall be finished smooth and free of burrs and sharp projections. Surface appearance shall be in accordance with the acceptable industrial practice.

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3.5 Safety. All rotating parts, except the chuck, and all live electrical parts shall be suitably guarded to prevent accidental contact by the operator.

3.6 Motor. The motor shall be of the continuous duty type and shall operate on direct current. The motor shall be manufactured in accordance with recognized electrical standards and suitable for the purpose intended.

3.6.1 Stalling of drill. Each drill shall be capable of being stalled ten times within one minute without damage or smoke.

3.7 Bearings, gears, and shafts. The bearings, gears, and shafts shall be suitable for the purpose intended and shall conform to and be applied in accordance with accepted standards. Gears shall be enclosed within a gear housing.

3.8 Lubrication. Drills shall be prelubricated by the manufacturer. Bearings and gears that require subsequent lubrication shall be arranged so that the proper lubricant can be applied. Pressure lubricating fittings shall not be used. Sufficient lubricant shall be put into the gear housing so that the gears will be constantly lubricated when in operation. The gear housing shall be suitably enclosed to prevent any leakage of the lubricant.

3.9 Switch. All drills shall be provided with a conveniently located switch for switching the current to the motor to an "on" and "off" position. The switch shall be the type which requires constant pressure to hold it in the "on" position.

3.10 Chuck. Each drill shall be provided with an industrial-rated, medium-duty (minimum) geared chuck and key, in the sizes shown in tables I and II.

3.11 Nameplate. Each drill shall be provided with a suitable metal nameplate, permanently marked and securely fastened to an integral part of the frame or the housing in a conspicuous location. The marking shall include the manufacturer's name, type, or model number, serial number, size of drill, type of current (d.c.), voltage, and no-load spindle speed in r.p.m.

3.12 Radio interference suppression. When specified (see 6.1), drills shall be radio interference suppressed in accordance with MIL-I-43121.

3.13 Type I drills. Type I drills shall conform to the requirements of table I and shall be designed to operate on 10–12 volts. As specified (see 6.1), number 1, 2, and 3 drills shall have an end handle with closed grip (see figure 1) or an end handle with pistol grip (see figure 2). Number 4 drills shall have a combination breastplate and spade handle (horizontal or vertical type), with the switch located in a side handle (see figure 3). Numbers 3 and 4 drills shall also be furnished with an auxiliary side handle similar to that shown on figure 1 or 3. Numbers 1 and 2 drills may be provided with a removable auxiliary side handle at the option of the supplier.

TABLE I. Type I drill requirements at 10–12 volts d.c.

Drill number	Drill size	Chuck size	Weight with cable assembly (Maximum)	Speed [1] of chuck with no load	Output at chuck (rated load) (Minimum)	Torque at chuck (Minimum)	
						At rated load speed	Stalling
	Inch	Inch	Pounds	R.P.M.	Horse-power	Foot-pounds	Foot-pounds
1	¼	0–¼	6.5	1500	.14	.87	2.75
2	⅜	0–⅜	7.0	1000	.18	1.26	7.00
3	½	0–½	7.5	750	.20	2.56	9.20
4	½	0–½	8.5	450	.24	5.60	19.50

[1] The no-load speed shall be within 10 percent of the values specified and the speed at rated load shall be not less than 55 percent of the no-load speed.

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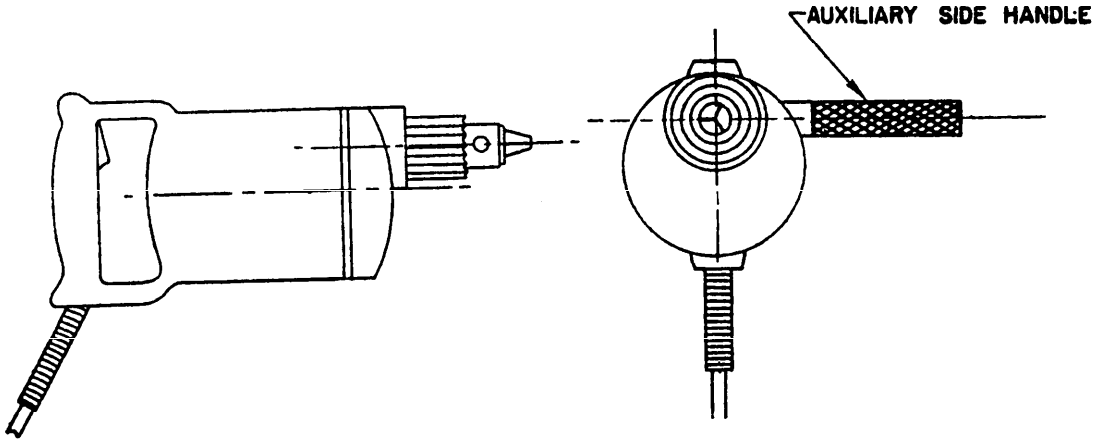


Figure 1. Type I drill with closed handle. (Also showing auxiliary side handle).

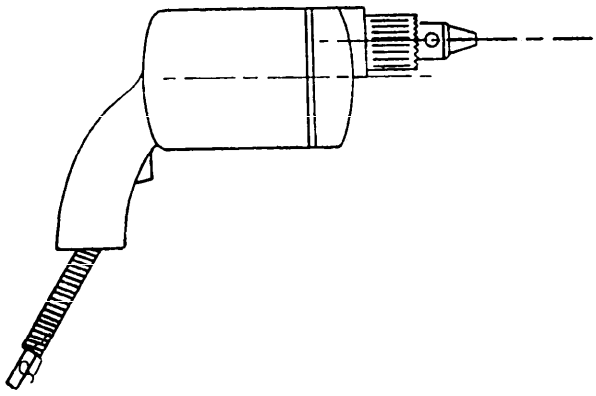


Figure 2. Type I drill with pistol grip handle.

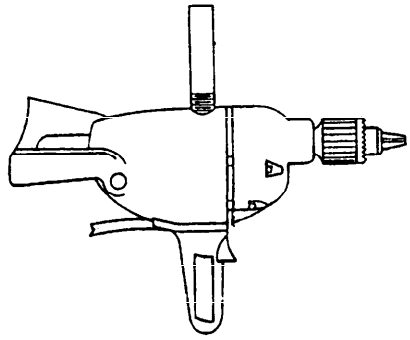


Figure 3. Type I drill with spade handle, side handle and auxiliary side handle.

3.13.1 Cable assembly. Each type I drill shall be provided with a cable assembly which is not less than 4 feet nor more than 5½ feet long. The cable assembly shall consist of a flexible cord and plug. The cord shall be equal to or exceed the requirements for type SJO or SJT construction, with two conductors of a size compatible with the current draw of the drill. The cord shall be reinforced at its junction with the drill for protection against sharp bends and shall have a strain relief to prevent mechanical stress on the cord from being transmitted to internal wiring and terminals. The plug shall be moulded to the cord and conform to the dimensions and polarity as shown on figure 4.

3.13.2 Accessories for type I drills. Unless otherwise specified in the contract or order (see 6.1), the following accessories are not furnished with the drill.

3.13.2.1 Battery for type I drills. Type I drills shall be designed to operate on a sealed nickel cadmium rechargeable 10 volt battery. The battery shall have a minimum of 4 ampere-hours rating and shall be capable of being recharged a minimum of 200 times. The battery shall have a receptacle for making electrical connection to the drill. The receptacle shall conform to the dimensions and polarity required to receive the plug connector specified in 3.13.1, as indicated on figure 4. The battery shall be provided with a suitable clip for attaching to the operator's waist belt. The battery shall weigh not more than 4½ pounds. The battery shall be capable of being charged with a battery charger which is in accordance with type II of W-C-265.

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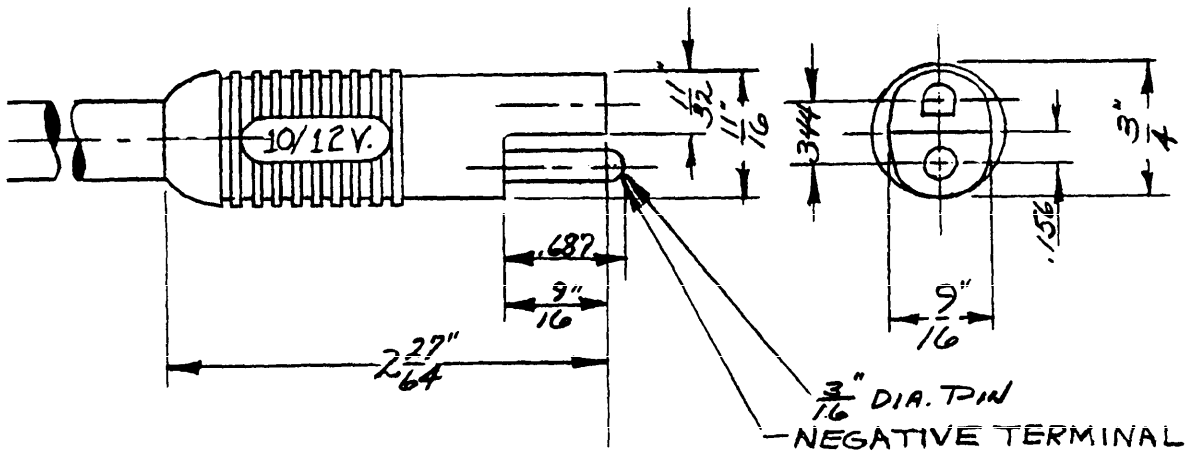


Figure 4. Dimensions and polarity for Type I drill plug connector.

3.13.2.2 Extension cable assembly. Extension cable assembly for type I drills shall be not less than 20 feet long. The assembly shall consist of a flexible cord and two plugs. The cord shall have two insulated conductors of size AWG 10 wire consisting of at least 103 strands of size AWG 30 copper wire each, suitable fillers, and a neoprene or thermoplastic jacket equal to or exceeding the requirements for type SJO or SJT construction. The plugs shall be moulded to the cord and be of proper design to connect the battery specified in 3.13.2.1 to the cable assembly specified in 3.13.1.

3.13.2.3 Adaptor cable assembly. Adaptor cable assemblies for type I drills shall be not less than one foot long. The assembly shall consist of a flexible cord, one plug and two spring tension electrical clips. The cord shall have two insulated conductors of size AWG 10 wire consisting of at least 103 strands of size AWG 30 copper wire. The plug shall be moulded to one end of the cord and be of proper design to connect to the cable assembly specified in 3.13.1. One clip shall be attached to each conductor so as to permit proper connection to a lead-acid storage battery with terminals one foot apart. The clips shall have a minimum jaw opening of one inch and be capable of conduction at least 50 amperes.

3.13.2.4 Type I accessory markings. All accessories shall be permanently marked with the manufacturer's name and part number. The battery shall be permanently marked with the voltage and instructions for charging.

3.14 Type II. The type II drills shall conform to the requirements of table II, and shall be designed to operate on the voltages as specified therein for the applicable size drill. The ¼- and ⅜-inch size drills shall have an end handle with pistol grip, similar to figure 5, which shall receive the battery. The ½-inch size drill shall have side handles which shall receive the batteries, and a two-position closed end handle (see figure 6). The ½-inch size drill shall be two-speed with a switch (marked with the speeds of "High" and "Low") for changing the speed, conveniently located on the drill housing. The drills shall also be provided with a conveniently located clip for holding the chuck key.

3.14.1 Screwdriver mechanism. When specified in the contract or order (see 6.1), size ⅜-inch, type II drills shall be provided with a suitable screwdriver mechanism and reversing switch for tightening and removing screws and nuts. The mechanism shall be an integral part of the drill, enclosed within the drill housing, and shall not affect the operation of the tool when used for drilling holes.

3.14.2 Accessories for type II drills. Unless otherwise specified in the contract or order (see 6.1), the following accessories are not furnished with the drill.

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TABLE II. Type II drill requirements at voltage indicated.

Drill size	Volts (d.c.)	Chuck size	Weight with battery (Maximum)	Speed [1] of chuck with no load	Output at chuck (rated load) (Minimum)	Torque at chuck (Minimum)	
						At rated load speed	Stalling
Inch		Inch	Pounds	R.P.M.	Horsepower	Foot-pounds	Foot-pounds
1/4	9	0–1/4	8	850	.07	.6	4.5
3/8	9	0–3/8	8	850	.07	.6	4.5
1/2	18	0–1/2	18	225	.16	5.5	27.0
				600		2.0	12.0

[1] The no-load speed shall be within 10 percent of the values specified and the speed at rated load shall be not less than 55 percent of the no-load speed.

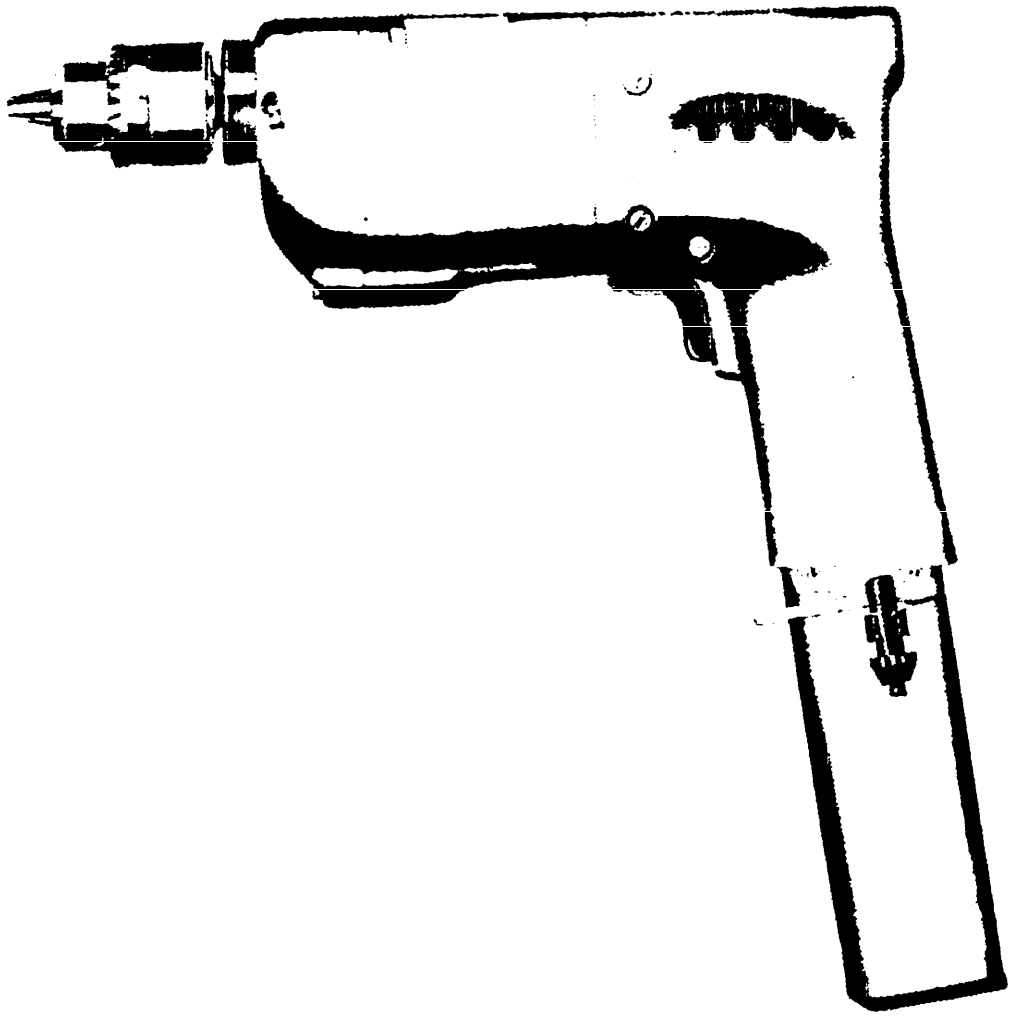


Figure 5. Type II drill, 1/4- and 3/8-inch size.

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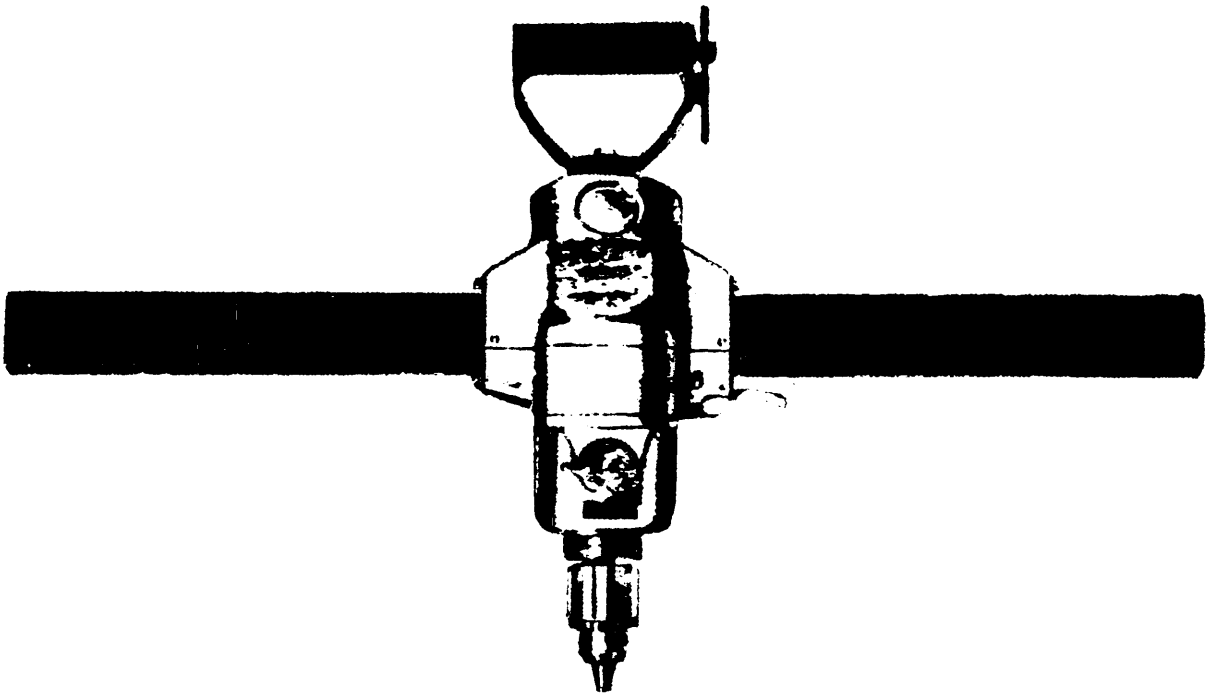


Figure 6. Type II drill, ½-inch size.

3.14.2.1 Battery for type II drills. Type II drills shall be designed to operate on 9-volt silver cadmium rechargeable batteries. The batteries shall have a minimum of 4.5 ampere-hours rating and shall be capable of being recharged a minimum of 200 times. The dimensions and connections shall be as shown on figures 7 and 8. The battery shall weigh not more than 2¼ pounds. Sizes ¼ and ⅜ inch drills require one battery and size ½ inch drills require two batteries for operation. The battery shall be capable of being charged with a battery charger which is in accordance with type I of W-C-265.

3.14.2.2 Type II accessory markings. All accessories shall be permanently marked with the manufacturer's name and part number. The battery shall be marked with the voltage, polarity, and instructions for charging.

3.15 Instruction book. One copy of an instruction book, consisting of the manufacturer's standard commercial instructions and parts list, shall be furnished with each drill.

3.16 Workmanship. Workmanship shall be of such quality to assure procurement of high quality equipment of a neat appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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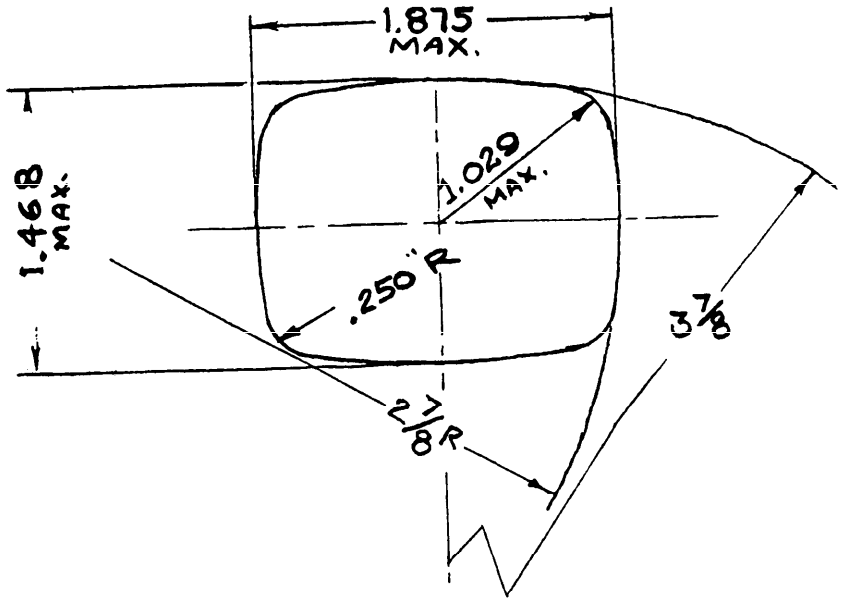


Figure 7. Dimensions for type II drill batteries.

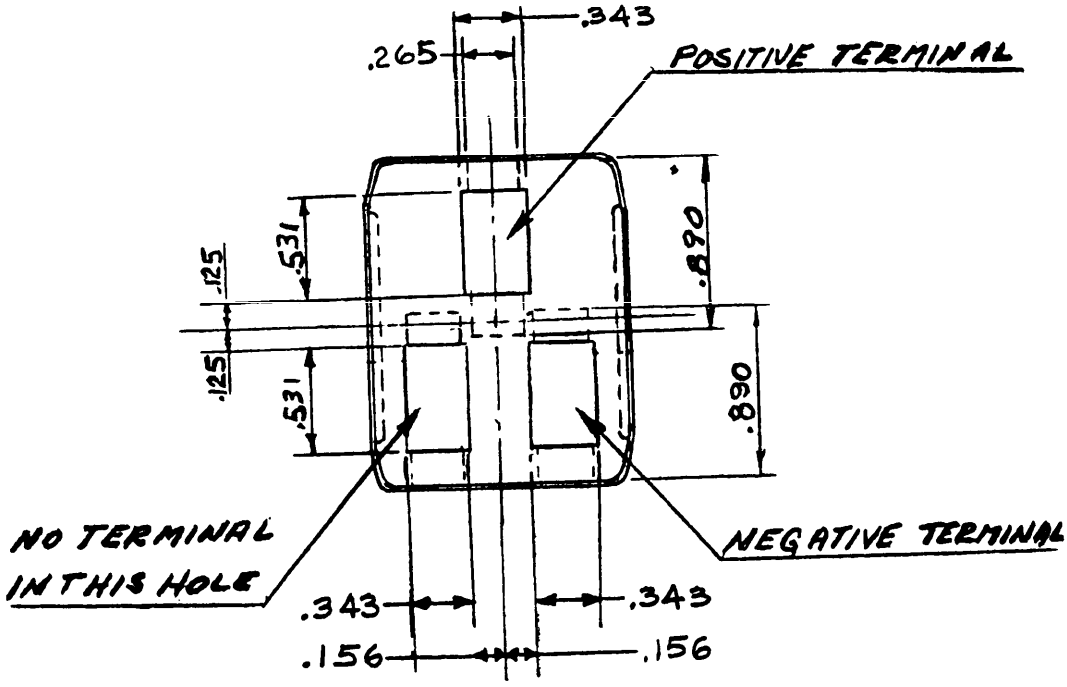


Figure 8. Terminal dimensions and polarity for type II drill batteries.

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4.2 Sampling for lot acceptance.

4.2.1 Lot. All drills or accessories of the same type and size, submitted for acceptance at one time, shall be considered a lot for purposes of inspection.

4.2.2 Sampling for examination. A random sample shall be selected from each lot for examination in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 Sampling for acceptance tests. A random sample shall be selected from each lot for acceptance tests in accordance with MIL-STD-105 at inspection level S-2. The AQL shall be 4.0 percent defective.

4.3 Inspection. Each of the sample drills or accessories selected in accordance with 4.2.2 shall be visually and dimensionally examined to determine compliance with this specification. Any drill or accessory in the sample containing one or more defects shall be rejected and if the number of defective drills or accessories in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

4.4 Acceptance tests. Each sample drill or accessory selected in accordance with 4.2.3 shall be tested in accordance with 4.5 and 4.6 to verify compliance with this specification. Any drill or accessory in the sample which does not meet the requirements for any of these tests shall be rejected, and if the number of nonconforming drills or accessories exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 Drill stall test. Each sample drill shall be connected to a constant source of direct current of its rated voltage and stalled ten times within one minute. The drill shall be rejected if any damage or smoke results.

4.5.2 Performance test. Each sample drill shall be connected to the appropriate battery or batteries (see 3.13.2.1 and 3.14.2.1) and tested for conformance with the applicable table (see tables I and II) for the following:

(a) Speed of chuck (at no load and at rated load).

(b) Output at chuck.

(c) Torque at chuck.

The drill shall be rejected if it does not meet the applicable requirements.

4.5.3 Accessory test. Each sample accessory shall be tested for compliance with this specification.

4.5.4 Radio interference suppression. Sample drills shall be tested to determine conformance with 3.12, when applicable.

4.5.5 Screwdriver mechanism. Sample drills shall be tested to determine conformance with 3.14.1, when applicable.

4.6 Endurance test. One drill from each lot shall be subjected to an 80-hour endurance test wherein the drill shall be operated at its rated load. The test shall be run in accordance with the manufacturer's recommended methods, but in no case shall the test time required exceed three weeks. Any indication of failure, during the test, shall be cause for rejection of the lot from which the drill was selected. If, at the time of invitation for bid, the manufacturer can present certified proof that the 80-hour endurance test has been successfully run on the item being offered, this test will not be necessary.

4.7 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2, and the AQL shall be not more than 4.0 defects per hundred units.

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TABLE III. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior containers)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as: Incomplete closure of container, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Packaging	Packages improperly sealed: Tape not as specified, joints not waterproof sealed.
	Cushioning improperly applied or missing.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Each type I drill with cable assembly or each type II drill without battery shall be packaged in a box conforming to PPP-B-636, V3c, style RSC. Die cut forms of built up layers of corrugated board shall be used to cushion the drill within the box. Each box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification. The batteries shall be packaged in a box conforming to PPP-B-665, style optional; or to PPP-B-676, type optional; or to PPP-B-566, style and type optional. Each box shall be securely closed to prevent accidental opening and waterproofed in accordance with MIL-P-116, method 2.

5.1.2 Level B. The drills and batteries shall be packaged as specified in 5.1.1, except that the boxes shall not be waterproof sealed with tape or waterproofed in accordance with MIL-P-116. The boxes shall be closed in accordance with the applicable specification.

5.1.3 Level C. The drills and batteries shall be preserved and packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Drills. Drills of the same type and size shall be packed in close-fitting boxes conforming to PP-B-621, class 2, style optional; or to PPP-B-601, overseas type, style optional; or to PPP-B-636, V3c, style RSC. The gross weight of the wood boxes shall not exceed 125 pounds; of the fiberboard boxes, the weight limitations of the box specification.

5.2.1.2 Batteries. The batteries shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style RSC. The gross weight of each box shall not exceed the weight limitations of the box specification. Each box shall be closed and waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B. The drills and batteries shall be packed as specified in 5.2.1, except that the boxes shall be domestic class and type and waterproofing the fiberboard boxes shall not be required.

5.2.3 Level C. The drills and batteries, when furnished, shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

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5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order (see 6.1), the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order (see 6.1), the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Types and sizes of drills required (see 1.2.1, 3.13, and 3.14).
(c) If radio interference suppression is required (see 3.12).
(d) Type of handle required for type I drills (see 3.13).
(e) Accessories required (see 3.13.2 and 3.14.2).
(f) If screwdriver mechanism is required for type II drills (see 3.14.1).
(g) Preservation, packaging and packing requirements (see 5.1).
(h) Marking required (see 5.3).

6.2 Battery powered drills are designed to operate from power sources other than those specified herein. Suitable adaptors and accessories are available from drill manufacturers.

Military Custodians:
Army—GL
Navy—SH
Air Force—69

Review activities:
Army—GL, EL
Navy—SH
Air Force—69

User activities:
Army—WC
Navy—YD

Military coordinating activity:
Navy—SH
(Project 5130-0113)

Civilian Interests:
GSA-FSS
CON-NBS
VA

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